Notice that they only extend up till the dashed line which is closest to the value of the analog signal at that instant (discrete values). In common video jargon, this is the frame rate of the audio. An inadequate sampling frequency leads to distortion and noise. Needless to say, larger values of bit depth and sampling frequency lead to better quality audio. If the Illustrative Graph was an actual analog to digital conversion, it would result in extremely poor quality audio as the samples are infrequent, and the bit values are few and far apart. For a more realistic (but still not very good quality) example, take a look at the 4-bit sampling graph on the previous page.

Nyquist-Shannon Sampling Theorem

Remember when we said that an 'adequate' sampling rate is required for effective analog to digital conversion? That is intuitively obvious, but this fundamental theorem lets us know exactly how much is 'adequate'. The Sampling Theorem, as it is commonly known, provides us the conditions necessary for A/D conversion such that no information is lost. This only applies to a finite bandwidth, but nature has taken care of that for us by putting limits on our hearing. The average human ear can hear from 20Hz to 20KHz so that is our bandwidth. According to the Sampling theorem, we can safely and reliably encode any analog signal within 20KHz as long as we use a sampling rate that is 40KHz or more. Of course there is more to the theorem than that, and it also provides a formula for reconstructing the analog wave from samples, but the key takeaway is that the sampling rate (Nyquist rate) should ideally be double the bandwidth limit (Nyquist Frequency) of the analog signal. Have a look at this page for an intuitive understanding: *http://dgit.in/NyqSThrm*

A link for the mathematically inclined: *http://dgit.in/SignSys*

Digital to Analog (D/A) conversion

DAC are electronic devices that perform the reverse function of ADCs. ADCs are only required while recording the audio but once it is stored digitally, it can be easily manipulated, enhanced, and most importantly, copied and transferred across the internet. DAC chips are present in every audio playback device that is capable of playing digital audio, from your mobile phone to your console.

A DAC takes as input a series of finite precision values and outputs a continuous time varying signal (which is usually voltage). In practice, the output of the DAC is a step function and a reconstruction filter is applied to it, to return it to its original analog form.